QUANTSIM PRO

Monte Carlo Simulation Engine

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Complete User Guide

Step-by-step instructions — from data to simulation results

Model
Geometric Brownian Motion (GBM)

Data Source
Yahoo Finance / Stooq / Investing.com CSV

Version
1.0 — March 2026

Please read the disclaimer before proceeding.
For developer details please refer to Section 8 on page number 14.

Disclaimer

This platform is a quantitative simulation tool for educational and informational purposes only.
It does not constitute investment advice, financial advice, trading advice, or any other form of recommendation.

Users are solely responsible for their investment decisions.

The simulations are based on historical data and mathematical models which rely on assumptions that may not hold in real-world markets. Actual market behavior may differ significantly.

Financial markets are subject to risk and uncertainty. Past performance and simulations are not indicative of future results.

We shall not be liable for any financial losses incurred based on the use of this platform.

Data is sourced from third-party providers and may contain errors, delays, or inaccuracies.

QuantSim Pro does not collect, store, or process any personal data.
All computations are performed locally within your browser, and no user inputs or results are transmitted to any server.

We do not require login, do not track user activity, and do not use cookies for data collection purposes.

Any market data used within the platform is fetched from third-party public sources solely for simulation purposes.

1. What is QuantSim Pro?

QuantSim Pro is a browser-based Monte Carlo simulation engine for financial assets. It takes real historical price data for any stock, commodity, index, or currency pair, calibrates a mathematical model from that data, and then simulates thousands of possible future price paths.

The output is not a single price prediction. It is a full probability distribution of outcomes — showing you the realistic range of where the price could be after any chosen number of trading days, with probabilities attached to every outcome.

1.1 What problem does it solve?

Standard financial analysis gives you a single forecast: "the stock will be at ₹500 in 3 months." This is misleading because markets are uncertain. QuantSim Pro gives you the full picture:

- There is a 19% probability the stock is above ₹500
- There is a 5% probability it falls below ₹380 (your VaR floor)
- The middle 50% of outcomes land between ₹420 and ₹530

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1.2 The mathematical model: Geometric Brownian Motion

The simulation uses GBM — the same model underlying the Black-Scholes option pricing formula. Each simulated price path follows:

$$S(t + dt) = S(t) \times \exp[\ (\mu - \tfrac{1}{2}\sigma^2) \times dt + \sigma \times \sqrt{dt} \times Z\ ]$$

Where:

- S(t) is the stock price at time t
- μ (mu) is the annual drift — the average direction the stock moves
- σ (sigma) is the annual volatility — how wildly the stock swings
- dt is one trading day = 1/252 of a year
- Z is a random draw from the standard normal distribution

Both μ and σ are automatically calculated from your historical data. Z is randomly sampled thousands of times to generate thousands of different possible futures.

2. How to Open the App

QuantSim Pro is a single .html file. It requires no installation, no login, and no internet connection to run (only the auto-fetch step needs internet).

1. Download the file: QuantSim_Pro.html
2. Double-click the file — it opens in Chrome, Firefox, or Edge
3. The simulator loads instantly in your browser

3. The 5-Step Workflow

The app guides you through 5 sequential steps using the step bar at the top of the left panel. Each step must be completed before moving to the next.

1 Step 1 — Fetch Market Data

This is where you tell the app which asset to analyse and how much historical data to use for calibration.

3.1.1 Entering a ticker symbol

Type the ticker symbol in the input box at the top. The app uses the following formats:

Asset type	Correct ticker format
Indian NSE stock	TCS.NS, RELIANCE.NS, JUBLFOOD.NS, INFY.NS
Indian BSE stock	TCS.BO, RELIANCE.BO
US stock	AAPL, TSLA, MSFT, GOOGL
Brent crude oil	BZ=F
WTI crude oil	CL=F
Gold	GC=F
Silver	SI=F
NIFTY 50 index	^NSEI
SENSEX index	^BSESN
S&P 500 index	^GSPC
NASDAQ 100	^IXIC

TIP Common short names are auto-resolved. You can type `TCS` and the app automatically converts it to `TCS.NS` before fetching. Similarly, `NIFTY` → `^NSEI` and `BRENT` → `BZ=F`.

3.1.2 Setting the data period

Use the slider to choose how many months or years of historical price data to use. Toggle between Months and Years using the segment control above the slider.

- Minimum recommended: 18 months (at least 378 trading days)
- Ideal: 3–5 years for most Indian stocks
- More data gives more reliable μ and σ estimates
- Be aware: very long periods may include regime changes (e.g. pre/post COVID)

3.1.3 Auto-fetch (7-source waterfall)

Click Fetch ↓ and the app attempts 7 data sources in sequence. The status bar below the button shows exactly which source is being tried in real time:

Attempt	Source
1	Stooq direct — fastest, no proxy, works in most browsers
2	Yahoo Finance direct — query1.finance.yahoo.com CSV endpoint
3	Yahoo Finance via corsproxy.io
4	Yahoo Finance via allorigins proxy
5	Yahoo query2 server via proxy
6	Stooq via allorigins proxy
7	Stooq via corsproxy.io

Each attempt has a 10–12 second timeout. If one fails, it moves to the next automatically. You do not need to do anything — just wait.

3.1.4 Manual CSV upload (fallback)

If all 7 sources fail (shown by a yellow Upload CSV button), you can download the CSV manually and upload it. The app reads all three common formats:

- Yahoo Finance CSV (Date, Open, High, Low, Close, Adj Close, Volume) — dates in YYYY-MM-DD
- Stooq CSV (Date, Open, High, Low, Close, Volume) — dates in YYYY-MM-DD
- Investing.com CSV (Date, Price, Open, High, Low, Vol., Change%) — dates in DD-MM-YYYY or MM/DD/YYYY

2 Step 2 — Confirm Base Price & Currency

You do not need to do anything in this step. The app automatically sets S_0 (the simulation starting point) to the last closing price in your historical data.

- If your data ends on 23-Mar-2026 with a closing price of ₹438.05, then S_0 = ₹438.05
- The simulation will project forward from this price
- All historical data before this date is used only for calculating μ and σ

The only thing to set here is the currency display:

Field	What to enter
Symbol (prefix)	The currency symbol that appears before the number: ₹ for INR, $ for USD, £ for GBP, € for EUR
Unit (suffix)	Optional unit that appears after the number. Leave blank for stocks. Use /bbl for crude oil, /oz for gold.

3 Step 3 — Calculated Parameters

All three model parameters are computed automatically from your historical data. This page shows you what was calculated and lets you adjust if needed.

3.3.1 Annual drift μ (mu)

This is the average annualised return of the stock based on its historical price movement.

- Calculated as: mean of daily log-returns × 252
- A positive μ means the stock has historically trended upward on average
- A negative μ means the stock has trended downward
- Example: Jubilant FoodWorks shows μ = −12.32% (the stock has lost ~12% per year over 2022–2026)

μ alone does not make GBM a good predictor — it is an average of a noisy process. Its main role is to set the drift direction of simulated paths.

3.3.2 Annual volatility σ (sigma) — the most important parameter

This measures how wildly the stock price swings, annualised. It is the primary driver of how wide the simulation fan is.

- Calculated as: standard deviation of daily log-returns × √252
- Higher σ = wider fan, larger VaR risk, larger upside potential
- Lower σ = narrower fan, more predictable outcomes

Asset type	Typical σ range
Large-cap Indian stock (TCS, Reliance)	20–35%
Mid-cap Indian stock (Jubilant, Zomato)	30–50%
US large-cap stock (AAPL, MSFT)	18–28%
Brent crude oil	25–45%
Gold	12–20%
NIFTY 50 index	15–25%
Bitcoin	60–110%

WARN **Mixed date formats inflate σ:** Some Investing.com CSV files contain both DD-MM-YYYY and MM/DD/YYYY formats in the same file. QuantSim Pro detects this and processes rows in file order (never re-sorts) to avoid phantom price jumps that would make σ appear 3–4× too high.

3.3.3 Risk-free rate r

The return available from a zero-risk investment (government bonds). Auto-set by detected currency:

Currency	Auto-set rate	Benchmark
INR (₹)	6.50%	RBI repo rate
USD ($)	4.50%	US Federal Reserve rate
GBP (£)	5.25%	Bank of England base rate
EUR (€)	3.50%	ECB main refinancing rate
JPY (¥)	0.10%	Bank of Japan policy rate
AUD (A$)	4.35%	Reserve Bank of Australia rate

4 Step 4 — Define Scenarios

This is where Monte Carlo simulation becomes genuinely powerful. You can run up to 4 different what-if scenarios simultaneously, each with a different volatility assumption. All scenarios share the same μ and S_0 but differ in σ.

3.4.1 How scenarios work

Each scenario applies a multiplier to σ (sigma):

- σ × 1.0 = Base case: normal market using exact historical volatility
- σ × 1.5 = Stress: elevated uncertainty (geopolitical tension, earnings risk)
- σ × 2.0 = Extreme stress: crisis conditions (COVID-level volatility, regulatory shock)
- σ × 0.7 = Calm: reduced volatility (stable macro, low-volume period)

3.4.2 Naming your scenarios

Give each scenario a descriptive name. Examples:

- "Base case — no conflict"
- "India–Pakistan border tensions (σ × 1.5)"
- "US tariff shock (σ × 2.0)"
- "Post-budget stability (σ × 0.8)"

The Effective σ field updates live as you change the multiplier, showing you the actual annualised volatility that will be used in the simulation.

TIP

Case competition strategy: Define your scenarios to match the specific risk factors in your case. For a crude oil case: Scenario 1 = historical baseline, Scenario 2 = Iran supply shock (σ × 1.5), Scenario 3 = OPEC+ production cut stabilisation (σ × 0.9). The delta section in the results will then show the exact dollar impact of each scenario on your VaR.

5 Step 5 — Run the Simulation

3.5.1 Setting the horizon

The horizon is how many trading days forward to simulate. This is the core question: "Where could this stock be after X trading days from S_0?"

Duration	Trading days
1 month	21 days
3 months	63 days (recommended starting point)

6 months	126 days
1 year	252 days
2 years	504 days

3.5.2 Number of simulations

This controls how many independent random price paths are generated. More paths = more accurate statistics but slower computation.

- 1,000 — Fast preview, visible sampling noise in tails
- 5,000 — Balanced, good for quick analysis
- 10,000 — Recommended: statistics stabilise, computation takes ~2-3 seconds
- 25,000 — High precision: very stable VaR and tail estimates

For most use cases, 10,000 is the optimal choice. The improvement in accuracy from 10,000 to 25,000 is marginal (the standard error scales as $1/\sqrt{N}$).

3.5.3 The simulation summary

Before clicking Launch, review the summary box which confirms all your settings: ticker, S_0, μ, σ, horizon, path count, and scenario count. Click Launch simulation — the engine runs entirely in your browser.

4. Reading the Results

4.1 The fan chart

Each scenario produces a fan chart showing all simulated paths collapsed into percentile bands:

Element	What it shows
Bright solid line	Median (p50) — the midpoint of all paths. Half ended above, half below.
Dark band	p25–p75: the middle 50% of all outcomes. This is where the stock spends most of its time.
Light band	p5–p95: the middle 90% of outcomes. This is your likely range under normal conditions.
Dashed line	Starting price S_0 — reference line to see profit/loss vs today.
Hover tooltip	Shows exact p5, p25, median, p75, p95 values at any trading day.

The fan always widens over time because uncertainty compounds. This is mathematically correct:

The further into the future, the less predictable the outcome.

4.2 The six metric cards

Below the fan chart, six cards summarise the distribution of all final prices at the horizon:

Metric	Definition and how to use it
Mean final price	The arithmetic average of all N simulated final prices. Under GBM this equals $S_0 \times e^{(\mu T)}$. A positive drift means mean > S_0 even when median < S_0 (due to log-normal asymmetry).
Median (p50)	The midpoint — 50% of simulations ended above this, 50% below. For a bearish stock, the median will be below S_0 even if the mean is above (because log-normal distributions are right-skewed).
VaR 95% — p5	Value at Risk: in the worst 5% of simulated scenarios, the price falls to at least this level. Use this as your downside risk floor when setting stop-losses or hedging coverage.
Upside 95% — p95	In the best 5% of simulated scenarios, the price reaches at least this level. Use this to size upside targets for options strategies.

Std deviation	The standard deviation of all final prices. Shows how dispersed outcomes are at the horizon. Larger = more spread, more risk and more opportunity.
Probability above S_0	The fraction of all 10,000 paths that ended above the starting price. **This is your probability of profit for a long position over the chosen horizon.**

4.3 The comparison table (multi-scenario only)

When you run 2 or more scenarios, a side-by-side table appears comparing all metrics across all scenarios. This makes it immediately clear how much additional risk or opportunity each scenario adds vs the base case.

The table includes 11 rows: σ used, mean, median, std dev, p5, p10, p25, p75, p90, p95, probability above S_0, probability of +10% gain, probability of −10% loss.

4.4 The scenario delta callout

At the bottom of the results, a green callout box shows the exact differences between each stress scenario and the base case:

- Δ p5 (downside risk): How much the worst-case floor drops in currency terms. E.g. "Geopolitical stress adds ₹45 of downside at p5."
- Δ p95 (upside range): How much the best-case ceiling rises. Stress scenarios typically widen both tails.
- Δ Std deviation: How much additional price uncertainty the stress scenario introduces.

KEY

The asymmetry insight: In a stress scenario (σ × 1.5), the upside gain (Δ p95) is almost always larger in absolute terms than the downside loss (Δ p5). This is because stock prices follow a log-normal distribution — they cannot go below zero but can go arbitrarily high. This means increased volatility benefits option buyers more than it hurts them.

5. Practical Use Cases

5.1 Portfolio risk management

Upload your stock, run a 3-month (63-day) simulation at 10,000 paths. Read the p5 (VaR 95%). This is the price the stock could fall to with only a 5% probability — your 95% confidence downside floor. Set your stop-loss slightly above this level.

Run a second scenario at σ × 1.5 to stress-test your hedge coverage under elevated market conditions.

5.2 Event / earnings stress testing

Before a quarterly earnings release or major macro event, run two scenarios:

- Scenario 1: Historical σ (pre-event baseline)
- Scenario 2: σ × 2.0 (post-earnings volatility spike)

The delta section shows you in rupee/dollar terms how much your risk exposure expands. This quantifies the premium you should pay for protective options.

5.3 Commodity price risk

For companies exposed to crude oil, gold, or other commodities:

- Upload BZ=F (Brent) or CL=F (WTI) data
- Set unit to /bbl
- Run a base case + geopolitical stress scenario
- The p5 of the stress scenario shows your worst-case procurement cost

This is directly useful for refiners, airlines, fertiliser companies, and any business with significant commodity exposure.

5.4 Options strategy planning

The probability above S_0 metric is a model-derived estimate of your probability of profit for:

- Long call option: probability the stock ends above the strike at expiry
- Long put option: (100% − probability above S_0) = probability the stock ends below current price

Use this alongside the p95 upside to calibrate your strike selection. If p95 is ₹550 and the current price is ₹438, an OTM call at ₹500 still has a meaningful probability of being reached.

5.5 Case competition analysis

6. Parameter Reference

Parameter	Description
S_0	Base price — always the last closing price in your historical data. The simulation projects forward from here.
μ (mu)	Annual drift. Computed as mean daily log-return × 252. Can be positive (uptrend) or negative (downtrend).
σ (sigma)	Annual volatility. Computed as std dev of daily log-returns × √252. The primary risk parameter.
r	Risk-free rate. Auto-set by currency. Used to discount future payoffs. Does not affect path simulation directly.
dt	Time step = 1/252. Represents one trading day as a fraction of a year. Fixed internally.
N (simulations)	Number of simulated paths. Recommended: 10,000.
T (horizon)	Simulation horizon in trading days. 63 = 3 months, 252 = 1 year.
σ multiplier	Scenario-specific factor applied to the base σ. 1.0 = no change, 1.5 = 50% more volatile.
p5 / VaR 95%	The 5th percentile of final prices. In 95% of simulations the price ended above this.
p95	The 95th percentile of final prices. In 95% of simulations the price ended below this.

7. Frequently Asked Questions

Why does the mean differ from the median?

Because stock prices follow a log-normal distribution, not a normal one. Log-normal distributions are right-skewed: there is more room for extreme upside (price can multiply many times) than extreme downside (price cannot go below zero). This means the mean is always slightly higher than the median. The median is a more intuitive measure for most purposes.

My σ seems very high (e.g. 120%). What went wrong?

This is almost always caused by mixed date formats in the CSV. Some Investing.com files switch between DD-MM-YYYY and MM/DD/YYYY mid-file. If the parser re-sorts dates incorrectly, it creates phantom 20–40% single-day moves that inflate σ by 3–4×. QuantSim Pro fixes this by processing rows in file order and detecting the separator character (dash vs slash) per row. If you still see an unrealistic σ, check the first and last few rows of your CSV for date format consistency.

The auto-fetch is stuck on "Trying source X/7"

Each source has a 10–12 second timeout. With 7 sources this means up to 84 seconds in the worst case. This is rare — usually sources 1 or 2 succeed within a few seconds. If all 7 fail, the yellow Upload CSV button appears. Download the CSV manually from Yahoo Finance, Investing.com and upload it.

Does more historical data always give better results?

Not necessarily. More data gives more statistically reliable μ and σ estimates. However, μ in particular is very sensitive to the chosen time period — a stock that grew 40% annually from 2020–2022 then fell 15% annually from 2022–2024 will show a very different μ depending on which period you include. Volatility (σ) is more stable across time periods. As a rule: use 2–5 years for most analyses, and consider whether the historical period is representative of the conditions you are projecting into.

8. About the developer

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This project has been developed using AI-assisted tools to accelerate development, structuring, and documentation. The overall design, implementation, and validation of the model and its outputs have been carried out by the developer.

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